PRACTICAL 4: DISCONNECTED ARCHITECTURE (ADO.NET)

AIM: Design a webpage to demonstrate a disconnected architecture.

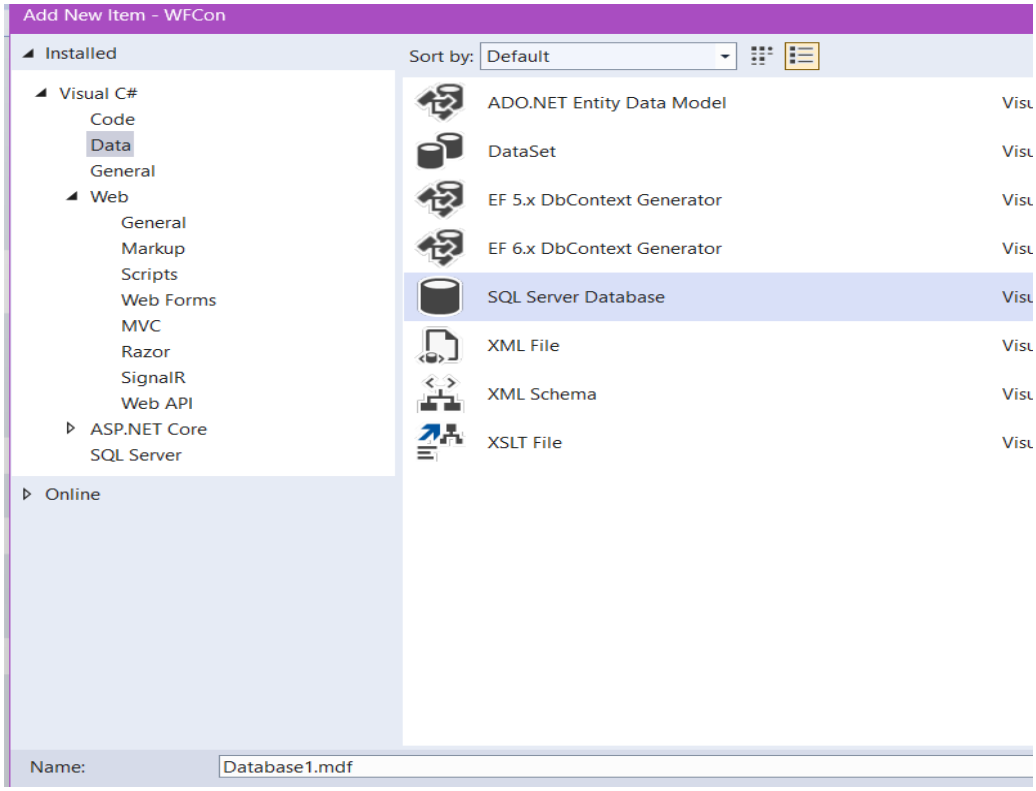
Step 1:

1. Open Visual Studio
2. Click **Create New Project**
3. Select **ASP.NET Web Application (.NET Framework)**
4. Click **Next**
5. Enter Project Name → DisconnectedDemo
6. Click **Create**
7. Select **Empty Template**
8. Tick **Web Forms**
9. Click **Create**

DATABASE PART

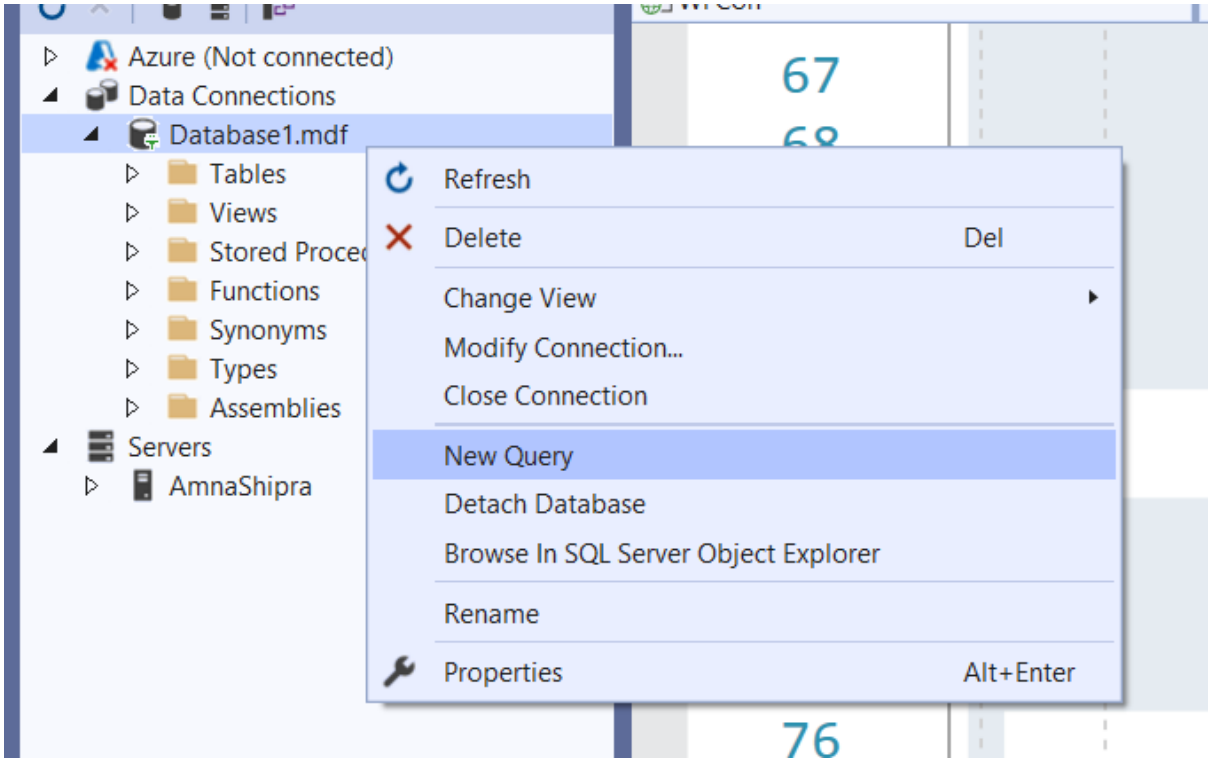
Step 2: Add Database

1. Right-click Project → **Add** → **New Item**
2. Select **SQL Server Database**
3. Name → Database1.mdf



Step 3: Create Table

Right Click on Database→ New Query



Run this query:

```
CREATE TABLE student
(
   sname varchar(100),
   sroll int Primary Key,
   sadd varchar(100),
   scont varchar(100)
)
```

Step 4: Insert Sample Data

INSERT INTO student VALUES ('Aisha','1','Mumbai','9999999999')

FRONTEND

Create Web Page

1. Right-click Project → **Add** → **New Item**
2. Select **Web Form**

Add Controls

TextBoxes:

- TextBox1 → Name
- TextBox2 → Roll
- TextBox3 → Address
- TextBox4 → Contact

Buttons:

- Button1 → Select
- Button2 → Insert
- Button3 → Update
- Button4 → Delete
- Button5 → Clear
- Button6 → Show All

Labels:

- Label5, Label6, Label7, Label8

GridView:

- GridView1

WebForm1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="DisconnectedDemo.WebForm1" %>

<!DOCTYPE html>
<html>
<head runat="server">
    <title>Disconnected Architecture</title>
    <style type="text/css">
        .auto-style2 {
            font-size: x-large;
        }
        .auto-style3 {
            font-size: x-large;
            font-weight: normal;
        }
    </style>
</head>
<body>
<form id="form1" runat="server">

<h2 class="auto-style3">Employee Form (Disconnected Architecture)</h2>

    <span class="auto-style2">Emp No: </span> <asp:TextBox ID="TextBox1"
runat="server" CssClass="auto-style2"></asp:TextBox><br class="auto-style2" /><br
class="auto-style2" />
```

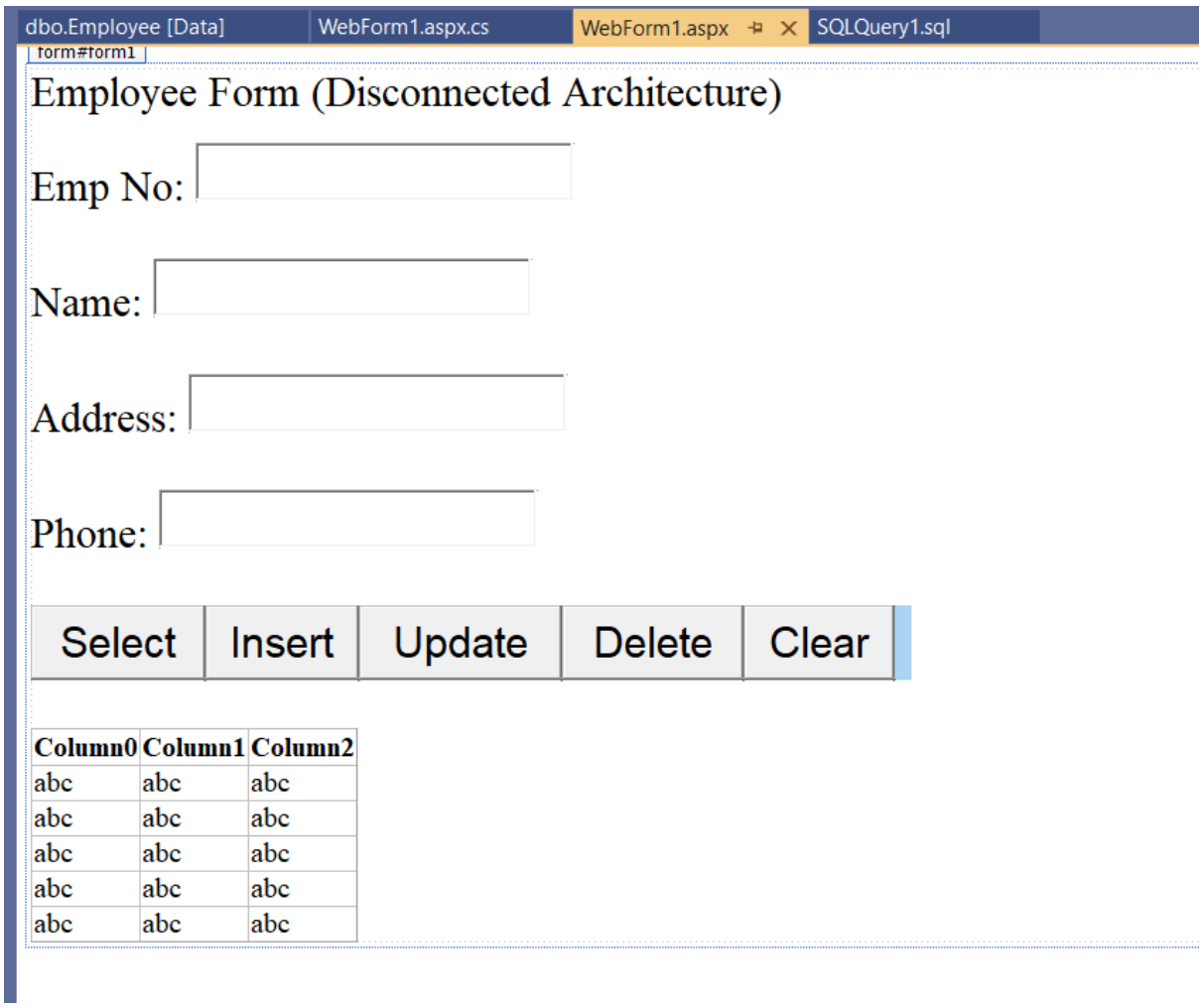
```
    <span class="auto-style2">Name: </span> <asp:TextBox ID="TextBox2" runat="server"
CssClass="auto-style2"></asp:TextBox><br class="auto-style2" /><br class="auto-style2"
/>
    <span class="auto-style2">Address: </span> <asp:TextBox ID="TextBox3"
runat="server" CssClass="auto-style2"></asp:TextBox><br class="auto-style2" /><br
class="auto-style2" />
    <span class="auto-style2">Phone: </span> <asp:TextBox ID="TextBox4" runat="server"
CssClass="auto-style2"></asp:TextBox><br class="auto-style2" /><br class="auto-style2"
/>

<asp:Button ID="Button1" runat="server" Text="Select" OnClick="Button1_Click"
CssClass="auto-style2" />
<asp:Button ID="Button2" runat="server" Text="Insert" OnClick="Button2_Click"
CssClass="auto-style2" />
<asp:Button ID="Button3" runat="server" Text="Update" OnClick="Button3_Click"
CssClass="auto-style2" />
<asp:Button ID="Button4" runat="server" Text="Delete" OnClick="Button4_Click"
CssClass="auto-style2" />
<asp:Button ID="Button5" runat="server" Text="Clear" OnClick="Button5_Click"
CssClass="auto-style2" />

<br class="auto-style2" /><br class="auto-style2" />

<asp:GridView ID="GridView1" runat="server"></asp:GridView>

</form>
</body>
</html>
```



Backend Code:

WebForm1.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;

namespace DisconnectedDemo
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        SqlConnection con;
        SqlDataAdapter da;
        SqlCommand cmd;
        DataSet ds;

        protected void Page_Load(object sender, EventArgs e)
        {
            con = new SqlConnection(
            @"Data Source=(LocalDB)\MSSQLLocalDB;
              AttachDbFilename=|DataDirectory|\Database1.mdf;
              Integrated Security=True");
        }

        // SELECT (DISCONNECTED)
        protected void Button1_Click(object sender, EventArgs e)
        {
            con.Open();

            da = new SqlDataAdapter("select * from Employee", con);
            ds = new DataSet();

            da.Fill(ds);

            con.Close();

            GridView1.DataSource = ds;
            GridView1.DataBind();
        }

        // INSERT
        protected void Button2_Click(object sender, EventArgs e)
        {
            string qry = "insert into Employee values(" +
                         TextBox1.Text + ",'" +
                         TextBox2.Text + "','" +
                         TextBox3.Text + "','" +
                         TextBox4.Text + "')";

            con.Open();

            cmd = new SqlCommand(qry, con);
            cmd.ExecuteNonQuery();

            con.Close();

            Response.Write("Inserted Successfully");
        }

        // UPDATE
        protected void Button3_Click(object sender, EventArgs e)
        {
            string qry = "update Employee set " +
                         "empname='" + TextBox2.Text + "'," +
                         "empaddress='" + TextBox3.Text + "'," +
```

```
                        "empphoneno='" + TextBox4.Text + "' " +
                        "where empno=" + TextBox1.Text;

            con.Open();

            cmd = new SqlCommand(qry, con);
            cmd.ExecuteNonQuery();

            con.Close();

            Response.Write("Updated Successfully");
        }

        // DELETE
        protected void Button4_Click(object sender, EventArgs e)
        {
            string qry = "delete from Employee where empno=" + TextBox1.Text;

            con.Open();

            cmd = new SqlCommand(qry, con);
            cmd.ExecuteNonQuery();

            con.Close();

            Response.Write("Deleted Successfully");
        }

        // CLEAR
        protected void Button5_Click(object sender, EventArgs e)
        {
            TextBox1.Text = "";
            TextBox2.Text = "";
            TextBox3.Text = "";
            TextBox4.Text = "";
        }
    }
}
```

Employee Form (Disconnected Architecture)

Emp No: 1

Name: Amna

Address: Mumbai

Phone: 9221004243

Select Insert Update Delete Clear

empno	empname	empaddress	empphoneno
1	Amna		
2	qwer	qwert	1234567891
3	asd	thane	2345678090
4	asdq	Mira rd	4345678090

Employee Form (Disconnected Architecture)

Emp No: 1

Name: Amna

Address: Mumbai

Phone: 9221004243

Select Insert Update Delete Clear

empno	empname	empaddress	empphoneno
1	Amna	Mumbai	9221004243
2	qwer	qwert	1234567891
3	asd	thane	2345678090
4	asdq	Mira rd	4345678090